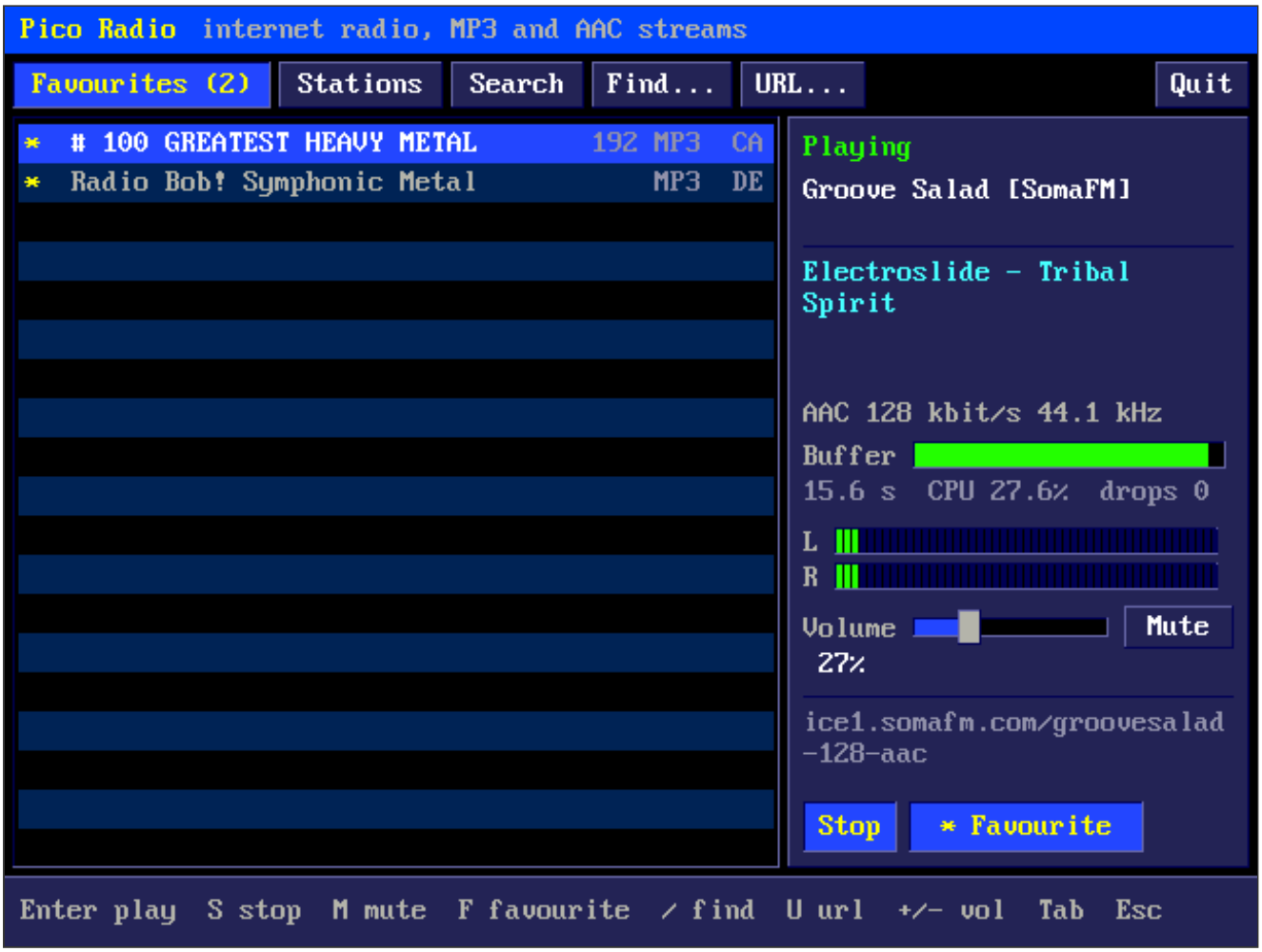
Pico 2 mini OS Internet Radio

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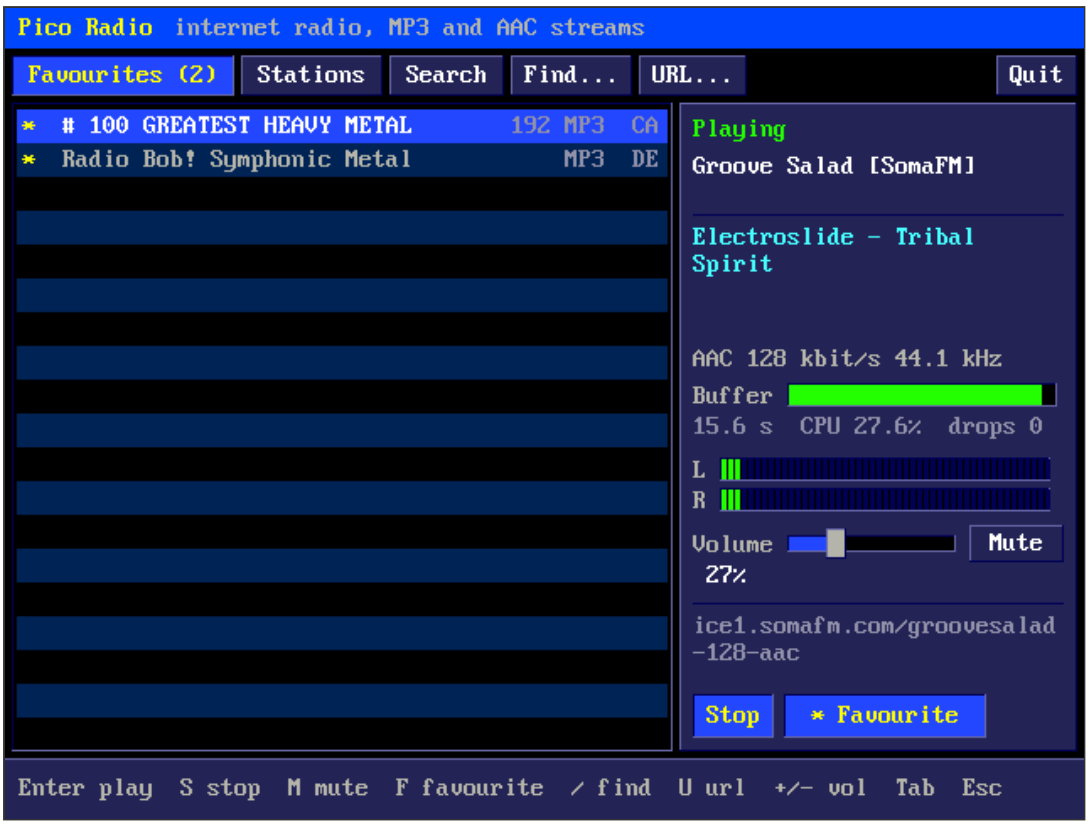


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Internet radio on the HDMI sound: MP3 and AAC streams over http and https, with a graphical station browser, a station search, favourites, and playing on in the background.

The radio



`radio` opens an internet radio in graphics: a list of stations, what is playing (the station, the song from the stream, bitrate, the buffer, level meters, the volume) and buttons, with the mouse or the keyboard. MP3 and AAC streams over http:// and https:// play, as Icecast and Shoutcast send them (with the song titles), through redirects and .pls/.m3u playlists. AAC means ADTS streams (audio/aac, audio/aacp). AAC-LC plays in full; HE-AAC (AAC+, usually 64 kbit/s or less) plays its AAC core only, at half the sample rate (22 or 24 kHz), so it sounds duller than on a PC: its SBR part needs about 50 KB more of fast memory than the system has to spare. The codec comes from the stream's Content-Type, or from its first frames when the type doesn't say, and shows in the lists and under the song (MP3, AAC, or HE-AAC for a stream at 24 kHz or less). Ogg, Opus, FLAC, MP4 and HLS streams aren't supported (a search lists only MP3 and AAC stations).

- **Lists:** Favourites, Stations (a built-in list: SomaFM, Radio Paradise, Radio Swiss, 181.FM) and Search: Find (or `/`) searches radio-browser.info by name, or `tag:jazz`, `country:DE`, `lang:french`, most listened first. URL... (or `U`) plays any address.
- **Keys:** Up/Down/PgUp/PgDn choose, Enter plays, S stops, F (or the star at the left of a row) adds or removes a favourite (Del in Favourites), +/- or Left/Right the volume, M mutes and unmutes (the stream goes on; MUTED shows under the volume, and the Mute button beside the slider does the same), `[ ]` turn the bass down and up and `{ }` the treble (2 dB a press, -12 to +12; the Bass and Treble sliders under the volume show them), Tab the next list, Esc leaves.
- **Bass and treble:** shelves at 150 Hz and 4 kHz on the 48 kHz sound. The bass starts at +6 dB, since the speakers in a TV or monitor give little; set it to 0 for flat on good

speakers. A boost first lowers the whole by half of it, so loud music isn't squashed by the limiter.

- **Mouse:** click a row, double-click (or right-click) to play, the wheel scrolls, drag the volume, bass and treble sliders (the tone ones are flat in the middle), click the tabs and buttons.
- **In the background:** leaving with Esc keeps the radio playing at the shell prompt and in other programs that don't use the sound (`radio stop` stops it; MUSIC, the tracker and the emulators take the sound over). `radio status` says what is playing, `radio last` plays the last station again, `radio play URL` plays an address, `radio volume 0-100`, `radio bass dB`, `radio treble dB`, `radio mute [on|off]` (alone: toggles). **Ctrl+Alt+M** (or a keyboard's Mute key) mutes and unmutes the radio from anywhere, in the shell, the editor, BASIC or any program, with a note at the top right for a moment.
- Favourites, the last station, the volume and the bass and treble are kept in `/RADIO.INI` (`fav=name|url|kbit/s|country|codec` lines, `volume=`, `bass=`, `treble=`; it can be edited).
- How it works: a worker in the network's own interrupt reads the stream (decrypting https) into a 256 KB buffer in PSRAM (about 16 s at 128 kbit/s; playing starts after about 2.5 s), taking the song titles out; core 1 decodes it with the Helix MP3 or AAC decoder (RealNetworks, fixed point, src/radio/helix and src/radio/aac) and resamples it to 48 kHz, then the bass and treble (src/radio/radio_tone.c; `tools/radiotest/tonetest.sh [song.mp3]` checks them on a PC). The decoder's buffers are in the apps' shared SRAM area while it plays (no heap); nothing stays allocated after `radio stop`.